Select catering systems

D1.HCA.CL3.07

Trainee Manual

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William Angliss Institute
Specialist centre for foods, tourism & hospitality

Select catering systems

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Trainee Manual

Project Base

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The Association of Southeast Asian Nations (ASEAN) was established on 8 August 1967. The Member States of the Association are Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand and Viet Nam.

The ASEAN Secretariat is based in Jakarta, Indonesia.

General Information on ASEAN appears online at the ASEAN Website: www.asean.org.

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Disclaimer

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Introduction to trainee manual

To the Trainee

Congratulations on joining this course. This Trainee Manual is one part of a 'toolbox' which is a resource provided to trainees, trainers and assessors to help you become competent in various areas of your work.

The 'toolbox' consists of three elements:

- A Trainee Manual for you to read and study at home or in class
- A Trainer Guide with Power Point slides to help your Trainer explain the content of the training material and provide class activities to help with practice
- An Assessment Manual which provides your Assessor with oral and written questions and other assessment tasks to establish whether or not you have achieved competency.

The first thing you may notice is that this training program and the information you find in the Trainee Manual seems different to the textbooks you have used previously. This is because the method of instruction and examination is different. The method used is called Competency based training (CBT) and Competency based assessment (CBA). CBT and CBA is the training and assessment system chosen by ASEAN (Association of South-East Asian Nations) to train people to work in the tourism and hospitality industry throughout all the ASEAN member states.

What is the CBT and CBA system and why has it been adopted by ASEAN?

CBT is a way of training that concentrates on what a worker can do or is required to do at work. The aim is of the training is to enable trainees to perform tasks and duties at a standard expected by employers. CBT seeks to develop the skills, knowledge and attitudes (or recognise the ones the trainee already possesses) to achieve the required competency standard. ASEAN has adopted the CBT/CBA training system as it is able to produce the type of worker that industry is looking for and this therefore increases trainees' chances of obtaining employment.

CBA involves collecting evidence and making a judgement of the extent to which a worker can perform his/her duties at the required competency standard. Where a trainee can already demonstrate a degree of competency, either due to prior training or work experience, a process of 'Recognition of Prior Learning' (RPL) is available to trainees to recognise this. Please speak to your trainer about RPL if you think this applies to you.

What is a competency standard?

Competency standards are descriptions of the skills and knowledge required to perform a task or activity at the level of a required standard.

242 competency standards for the tourism and hospitality industries throughout the ASEAN region have been developed to cover all the knowledge, skills and attitudes required to work in the following occupational areas:

- Housekeeping
- Food Production
- Food and Beverage Service

- Front Office
- Travel Agencies
- Tour Operations.

All of these competency standards are available for you to look at. In fact you will find a summary of each one at the beginning of each Trainee Manual under the heading 'Unit Descriptor'. The unit descriptor describes the content of the unit you will be studying in the Trainee Manual and provides a table of contents which are divided up into 'Elements' and 'Performance Criteria". An element is a description of one aspect of what has to be achieved in the workplace. The 'Performance Criteria' below each element details the level of performance that needs to be demonstrated to be declared competent.

There are other components of the competency standard:

- *Unit Title:* statement about what is to be done in the workplace
- *Unit Number:* unique number identifying the particular competency
- *Nominal hours:* number of classroom or practical hours usually needed to complete the competency. We call them 'nominal' hours because they can vary e.g. sometimes it will take an individual less time to complete a unit of competency because he/she has prior knowledge or work experience in that area.

The final heading you will see before you start reading the Trainee Manual is the 'Assessment Matrix'. Competency based assessment requires trainees to be assessed in at least 2 – 3 different ways, one of which must be practical. This section outlines three ways assessment can be carried out and includes work projects, written questions and oral questions. The matrix is designed to show you which performance criteria will be assessed and how they will be assessed. Your trainer and/or assessor may also use other assessment methods including 'Observation Checklist' and 'Third Party Statement'. An observation checklist is a way of recording how you perform at work and a third party statement is a statement by a supervisor or employer about the degree of competence they believe you have achieved. This can be based on observing your workplace performance, inspecting your work or gaining feedback from fellow workers.

Your trainer and/or assessor may use other methods to assess you such as:

- Journals
- Oral presentations
- Role plays
- Log books
- Group projects
- Practical demonstrations.

Remember your trainer is there to help you succeed and become competent. Please feel free to ask him or her for more explanation of what you have just read and of what is expected from you and best wishes for your future studies and future career in tourism and hospitality.

Unit descriptor

Select catering systems

This unit deals with the skills and knowledge required to Select catering systems in a range of settings within the hotel and travel industries workplace context.

Unit Code:

D1.HCA.CL3.07

Nominal Hours:

35 hours

Element 1: Establish enterprise requirements for a catering system

Performance Criteria

1.1 Research catering requirements the enterprise requires

1.2 Identify the enterprise constraints in selecting a system

Element 2: Evaluate catering systems

Performance Criteria

2.1 Identify a range of alternative catering systems

2.2 Evaluate agreed enterprise requirements against systems

Element 3: Recommend a catering system

Performance Criteria

3.1 Consider the advantages and disadvantages of systems in making recommendation

Assessment matrix

Showing mapping of Performance Criteria against Work Projects, Written Questions and Oral Questions

The Assessment Matrix indicates three of the most common assessment activities your Assessor may use to assess your understanding of the content of this manual and your performance – Work Projects, Written Questions and Oral Questions. It also indicates where you can find the subject content related to these assessment activities in the Trainee Manual (i.e. under which element or performance criteria). As explained in the Introduction, however, the assessors are free to choose which assessment activities are most suitable to best capture evidence of competency as they deem appropriate for individual students.

		Work Projects	Written Questions	Oral Questions
Element 1: Establish enterprise requirements for a catering system				
1.1	Research catering requirements the enterprise requires	1.1	1 – 8	1 – 4
1.2	Identify the enterprise constraints in selecting a system	1.1	9, 10	5 – 8
Element 2: Evaluate catering systems				
2.1	Identify a range of alternative catering systems	2.1	11 – 21	9 – 13
2.2	Evaluate agreed enterprise requirements against systems	2.1	22, 23, 24	14, 15
Element 3: Recommend a catering system				
3.1	Consider the advantages and disadvantages of systems in making recommendation	3.1	25 – 31	16 – 19

Glossary

Term	Explanation
A la carte	(French) From the card (the card being the menu)
CEO	Chief Executive Officer
CFO	Chief Financial Officer
COP	Code of Practice
FSP	Food Safety Plan (or Program)
HACCP	Hazard Analysis Critical Control Points
HR	Human Resources
KSC	Key Selection Criteria
MICE	Meetings, Incentives, Conferences, and Exhibitions
Menu item	A dish listed on the menu
Mise en place	(French) To put in place; to get things ready
Primary research data	Original research information; freshly generated information
RFT	Request For Tender
Re-thermalisation	Re-heating
QA	Quality Assurance
Qualitative research data	Information providing descriptions on a research topic; also called 'soft' data
Quantitative research data	Statistical data – numbers, figures, percentages, costs; also called 'hard' data
SOP	Standard Operating Procedure

Term	Explanation
Secondary research data	Information gained from researching existing data
Sous vide	(French) Under vacuum
USP	Unique Selling Point/Proposition

Element 1:
Establish enterprise requirements for a catering system

1.1 Research catering requirements the enterprise requires

Introduction

A primary requirement when seeking to identify the catering system for a venue is to research the catering requirements of the organisation.

This section presents a context for the unit, identifies foundation skills and knowledge, discusses research methods and topics, addresses the concept of research data and lists those who may be involved in the research process.

Context for the unit

Enterprises and venues

This unit is applicable to a variety of commercial businesses that produce and serve food including:

- Hotels, taverns and bars
- Restaurants and cafes
- Private, sporting and other clubs
- School, universities and other educational institutions
- Hospitals, hospices and aged care facilities
- Workplace cafeterias and canteens
- Military (defence force) catering
- Prisons
- Residential caterers
- In-flight and other transport catering
- Meetings, Incentives, Conferences/conventions, and Exhibitions (MICE) catering.

Unit focus

The focus is on evaluating and selecting an integrated production, distribution and service catering system to meet the food production needs of a catering organisation.

The main aims address:

- Determination of catering system requirements for an organisation
- Evaluation of operational aspects of different catering systems
- Selection of a catering system which suits the characteristics and needs of the organisation being considered.

You will have a need for this unit when:

- Called on to modify an existing food production and food service system in a business
- The opportunity arises to build and install a new catering system for a venue or organisation.

Target employees

The unit is aimed at:

- Senior managers – such as including executive chefs and catering managers
- Who operate with significant autonomy – that is, they can make decisions with little or no reference to others in the organisation
- Who are responsible for making a range of strategic management decisions – relating to the direction of the business and ways to attain the identified goals of the organisation.

Catering system – defined

'Catering system' refers to an overall food production and food service system where all components/elements are integrated into a cohesive, effective and efficient operation.

It may include options such as:

- 'Conventional' – a system where food is cooked fresh and served at the time
- 'Cook-chill' – where food is cooked and stored under refrigeration for short-term or long-term storage

- 'Cook-freeze' – where food is cooked and frozen for later re-thermalisation and service
- 'Commissary' – featuring transportation of pre-prepared food to satellite kitchens for re-heating and service
- Assemble-serve – where pre-prepared food is portioned, plated and served: no cooking or other processing is required.

Foundation skills

People who undertake the task of selecting catering systems should posses the following skills to underpin their research, evaluation and allied activities:

- Communication skills to consult on system requirements with key personnel – such as:
  - Other management-level personnel
  - Boards of Directors
  - Owners
  - Government agencies and authorities
  - Equipment/system providers and suppliers

- Critical thinking skills – to:
  - Analyse and evaluate all aspects of the organisation's catering operation
  - Select a catering system which best suits its characteristics and needs
- Initiative and enterprise skills – to:
  - Determine courses of action
  - Maintain motivation and achieve required objectives or outcomes
  - Integrate workplace needs with capacity and capability of various catering systems
  - Select a system with the best cost benefits
- High level of literacy skills to:
  - Read and interpret detailed product specifications for different catering systems
  - Research product options for and suppliers of catering systems
  - Read and interpret recipes and menus
- High level numeracy skills to:
  - Calculate wastage issues and impacts on profitability
  - Determine cost-benefit analyses
  - Review complex financial information
  - Calculate costs of production and costs for installing a new system
- Planning, self-management and organising skills to:
  - Access and sort information required to evaluate different catering systems
  - Coordinate a timely and efficient selection process
  - Organise personal work and research efforts
- Problem-solving skills to:
  - Identify organisational operational constraints
  - Select a system which complements operations

- Teamwork and interpersonal skills to:
  - Invite and coordinate the input of others in the organisation
  - Facilitate liaison
  - Encourage contributions from, and engagement with, others

- Communication skills to:
  - Enable questioning of others
  - Facilitate exchange of ideas and information
- Research skills to:
  - Investigate relevant topics
  - Capture information
  - Follow-up information as required.

Foundation knowledge

It is to be expected those who are involved with the selection of catering systems will have significant catering knowledge to use as a platform for determining workplace needs for a catering and analysing system options.

This base knowledge should address all the following:

- Methods of cookery – for all major food types, including preserved and packaged foods for various types of hospitality and catering organisations
- Comprehensive details of all food production processes for:
  - Receiving – of food into the premises
  - Undertaking *mise en place* – organisation of ingredients and equipment/utensils prior to preparing/ and producing food
  - Preparing food – ready for cooking or processing
  - Cooking – the application of a variety of cooking options to produce menu items
  - Post-cooking storage – of foods for service, display and refrigerated or frozen storage for later use
  - Reconstituting foods – returning dehydrated or concentrated food to usable condition, or its original state
  - Re-heating of previously cooked food – referred to as 're-thermalisation'
  - Serving food – for on-site eating or for take-away consumption
- Hazard and Critical Control Points (HACCP) – with reference to:
  - General principles and practices
  - Specific requirements of the Food Safety Plan/Program (FSP) as it applies to the host venue
- Culinary terms – commonly used in the industry related to food production systems
- Costing, yield testing and portion control in quantity food production

- Nutritional knowledge – as applicable to:
  - General food stuffs
  - Specific needs of the workplace for identified customer/target groups
- Local/host country legislation – as it applies to:
  - Food handling and food safety
  - Workplace safety and health
  - Industrial relations.

Research methods

'Research methods' refers to ways in which you can obtain the necessary information/data required to:

- Learn about catering system options
- Analyse/evaluate the options available
- Decide on the best system to meet the identified needs/requirements of your operation.

Standard research methods for obtaining necessary information/data include:

- Meeting with management of the venue to:
  - Confirm need for the process to select a catering system – and obtain necessary authorisations to proceed with the research
  - Identify plans they have for the future direction of the business – which are likely to impact the selection of a catering system.

    Future directions which may influence a decision could include:
    – Plans for expansion or contraction of the business
    – Plans to focus on an a new and different target market
    – Plans to re-position the venue in the marketplace
    – Plans to reduce or increase staff levels
  - Identify operational and acquisition constraints you are expected to operate under – see section 1.2
- Assessing published information on different catering systems by:
  - Visiting relevant websites
  - Reading relevant articles in industry magazines and journals
  - Reading reports from industry peak bodies and relevant government agencies and authorities
  - Reading product information brochures and equipment/system specifications

- Communicating with suppliers of catering systems and:
  - Providing information about the business and its plans and operational requirements
  - Seeking product or system information
  - Determining options available together with:
    - Price
    - Availability
    - Capacity
  - Having representatives visit and view your operation – so they gain first-hand knowledge of your venue and its requirements
- Discussing food production and service needs with colleagues to:
  - Identify enterprise and operational requirements – see 'Research topics' below
  - Determine preferences for catering system deliverables – and rationale for same
  - Involve relevant others in the project – and thereby generate interest, enthusiasm and commitment to the process and final selection

- Visiting other industry operators – and:
  - Viewing their facilities
  - Talking to operational staff who use the catering systems which are in place
  - Speaking with management about their impressions of effectiveness, efficiency and system performance and problems
- Viewing your current operation – and:
  - Inspecting and measuring the facilities
  - Talking to staff and watching staff at work
  - Noting bottle-necks and other problem areas
- Attending relevant industry 'food production and food service' meetings and events – such as
  - Conferences
  - Seminars
  - Product launches
  - Symposiums.

Research topics

A wide range of topics need to be researched to determine enterprise requirements for a catering system.

The research should address:

- Inputs to the system – ingredients, energy, equipment, labour and similar
- Outputs from the system – menu items, volume, timing, waste and similar.

This section lists issues which need to be addressed as part of the research process under important operational headings or concerns.

Nature of the operation

This should determine:

- Whether food is to be:
  - Produced and served at the same point
  - Produced in the one kitchen for service at multiple points/outlets in the same venue
  - Produced at a central kitchen for distribution for transportation or distribution to multiple points and outlets for service at these off-site locations – centralised food production using a number of supporting satellite kitchens to re-constitute and re-thermalise foods
- The general nature of the business – such as:
  - Hotel, fine dining
  - Work canteen
  - Institutional catering.

The menu

There is a need to identify:

- The type of menu – is it:
  - Rotating/cyclical, as is the case in most hospitals or institutions?
  - A la carte – featuring a need for cooked to order meals?
  - Buffet – requiring smorgasbord style service?
  - Functions menus – requiring quantity food production and service?
- Menu items being offered – for example:
  - Identification of dishes and recipes
  - Determination of cooking styles
  - Nomination of number of courses
  - The intentions of those who plan the menus for the venue
- The time of day the menu is for – such as:
  - Breakfast, lunch or dinner
  - Supper.

Production volume

This relates to demand for food and service and may be related to:

- Standard/normal trading and service requirements
- Demand at peak times
- Variations in demand on a sessional, daily, weekly or seasonal basis
- Impact of events on normal food production and service
- Nomination of projections ('number of meals served') for different trading times/occasions on, as appropriate to the organisation:
  - An hourly basis
  - A sessional basis
  - A daily basis.

Service areas

Research needs to identify:

- The areas from which service needs to occur – on- and off-site, as applicable: physical relationship between food production kitchen and service points – in terms of distance and time to travel
- Size of service area
- Facilities currently existing in service areas
- Transport, including vehicles and staffing, required to move food from production area to service points
- Legislated requirements and 'best practice' protocols for safe food transportation of hot, refrigerated or frozen food.

Storage and holding requirements

Research may be related to identifying:

- Hot and/or cold holding – of prepared food for display and/or service, in terms of:
  - Demand for (capacity) same
  - Type of equipment required – bain maries, warmers, cabinets
  - Location and type of existing facilities
- Amount of storage space required for:
  - Refrigerated storage – of raw and prepared foods
  - Frozen storage – of raw and prepared foods
- Type and capacity of existing storage facilities.

Nutritional and dietary requirements

Research should identify:

- If there is a need for certain nutritional requirements to be met for:
  - Certain dishes
  - Nominated outlets
  - Identified foods.
- Exactly what the nutritional requirements are – in terms of, for example:
  - Serve sizes
  - Vitamins
  - Energy
- Special need to cater for nominated dietary requirements as they apply, for example, to:
  - Health-related issues – such as:
    - Diabetic meals
    - Low-salt or low-fat meals
    - Allergy-related menu items
  - Cultural or religious requirements
  - Lifestyle preferences.

Relevant timeframes

Research should indicate:

- Opening times and trading hours of commercial premises – for all service points and outlets
- Scheduled service times for meals – in operations such as hospitals, prisons and schools
- Lead times – from ordering of raw ingredients/food from suppliers to delivery of food into the premises and kitchen stores
- Delivery and transportation times – from central kitchen to satellite kitchens and other service points.

Available space

Research may be related to:

- Whether the new or revised system is required to fit within an existing space – or if there is scope to expand production and service areas into additional space
- Details of current layout – of existing systems, equipment and facilities including utilities
- Amount of:
  - Additional space available for expansion
  - Space which is required to be saved as a result of the change or up-date to the catering system.

Customer requirements

Research should address:

- Defining and identifying 'customers' or consumers
- Describing their identified needs, wants and preferences
- Obtaining feedback from existing and potential target markets.

Ingredients purchased

In this regard research should address:

- The type, style, nature and form of the foods bought by the kitchen to produce menu items. This is relevant and important as:
  - It has implications for the equipment and staff needed to:
    - Store the food
    - Prepare the food
    - Process the food
  - It needs to align with other factors/issues – such as:
    - Required quality standards
    - Customer preferences
    - Value-for-money
    - Image and market position of the organisation.

Enterprise practices and standards

Research may be related to:

- Understanding options for buying different ingredient/raw materials – or pre-prepared and ready-made items
- Knowing the contents of public statements the business makes about itself – in terms of:
  - Mission statement
  - Vision statement
  - Value statement
- Understanding Standard Operating Procedures (SOPs) and relevant policies for the operation
- Knowing the kitchen (or individual service point/outlet) quality standards for:
  - Individual menu items
  - Food service to customers
- Identifying the capacity for change within the business – and:
  - The procedures to achieve such change
  - The resources available to enable/support such change

- Determining any strategic or competitive advantages enjoyed by the business – in terms of (for example):
  - Unique selling points – products and/or services that other businesses do not have
  - Information, experience and/or expertise
  - Industry knowledge
  - Internal systems
  - Supplier contracts
  - Industry contacts and/or partners.

Utilities

Research should address:

- Availability of:
  - Electricity
  - Gas
  - Water
- Continuity and reliability of supply
- Cost
- Access.

Research data

All research data can be classified as:

- Secondary or primary
- Qualitative or quantitative.

You must aim to obtain representative and relevant information in all of these classifications.

Secondary and primary data

Secondary data is data/information that already exists.

It can be obtained by:

- Reading reports, articles and books
- Reviewing internal business trading information, performance statistics and operational reports
- Manipulating existing data.

Primary data is information which is original data.

It is generated as a result of:

- Asking questions and talking to people
- Observing practice and operations
- Market research activities – such as surveys, questionnaires and focus groups.

Qualitative and quantitative data

Qualitative data, also known as 'soft' data, relates to:

- Descriptions of things – such as:
    - "A large venue with extensive facilities serving a wide variety of target markets"
    - "The kitchen features conventional service and uses equipment which was installed in the venue after being removed from the organisation's previous property in Singapore"
- Explanations of preferences and behaviour – detailing:
    - Why people have selected certain systems or equipment for their food production and service
    - Reasons why businesses do not use certain systems or technologies in their food production and service
    - What other organisations think about nominated food production and food service options
- Any issues which cannot be measured/quantified – essentially these issues revolve around answers to 'Why?' questions:
    - "Why did you do this?"
    - "Why did you *not* do that?"

Quantitative data, also known as 'hard' data, is statistical in nature.

It includes:

- Numbers/figures – relating to topics such as:
    - Costs – of equipment, installation and estimates for service, maintenance and repairs
    - Time – to deliver, install and commission systems
    - Speed – of processing food, delivering meals and cooking dishes
    - Temperature – of cooking and holding equipment
    - Demand levels – for meals
    - Capacity and volume – of individual items of equipment or systems for delivering or producing menu items
- Percentages – such as:
    - Food cost percentage – the amount represented by the cost of food in the selling price of a menu item
    - Labour percentage – the amount of labour wages in the selling price of a dish
    - Return on Investment – the profit the venue obtains based on money invested.

Those involved in the research

While it should be you who drives the overall selection process there will always be a need to involve, consult and liaise with others.

The exact range of these stakeholders will depend on the organisation you are working with/for and their individual organisational structure and job positions.

This said, the people you need to involve and work with as part of your research activities and decision making processes can include:

- Senior management – including:
  - Venue managers
  - Director or CEO
  - The Board of Management, Board of Directors or the Executive
  - Head Office representatives
- Owners
- Contractors and sub-contractors – especially where aspects of catering have been outsourced
- Accountants and financial management staff/CFO – including external lenders
- Specialist consultants – with expertise in food production and service

- Head chefs/Executive chefs and section chefs
- Food and Beverage managers/supervisors or Foodservice Directors
- Function, Event and Banquet managers
- Menu planners
- Dieticians and nutritionists
- A combination of management-level employees and operational staff from a range of internal departments/divisions and workplace groups/teams with responsibility for:
  - Sales and Marketing
  - HR
  - Purchasing
  - Training and Development
  - Food safety
  - Workplace health and safety
  - Maintenance
- Representatives from suppliers – who provide system elements (technology and equipment)
- Officers and inspectors from local health and food safety authorities
- Representatives from target customer group.

1.2 Identify the enterprise constraints in selecting a system

Introduction

In order to gain a proper and full understanding of enterprise requirements for a new or up-dated catering system it is necessary to identify the constraints under which the system must be selected.

This section presents and discusses a range of common issues which frequently impose limitations on the selection of a catering system and introduces the concept of Key Selection Criteria.

Financial constraints

In relation to financial constraints:

- There will always be limits on what you can spend – there is never total financial freedom
- It is vital to talk to management to determine the amount of money available for the project – they may be able to:
  - Move money between budgets to allocate more money
  - Raise more funds than originally allocated
- You may be required to acquire the new system in different ways due to short-falls in cash, lack of availability of credit or cash flow issues – for example, you may be required to:
  - Lease items or systems rather than purchase them outright
  - Seek financing from suppliers
  - Source funds from 'other' financial institutions or investor sources
  - Request grants or subsidies from government agencies/bodies
- You will always be required to align with internal finance-related policies and procedures – which can include:
  - Need to obtain multiple written quotations from different suppliers – rather than only getting one quote form one supplier
  - Need to negotiate price – as opposed to simply 'accepting' prices quoted
  - Need to negotiate terms of payment – which would address:
    - Delaying or deferring dates by which various payments such as deposits, initial payment, progress payments and final payment to suppliers have to be made
    - Reducing amount of deposit and/or other payments
  - Need to negotiate other contracted terms – such as negotiating more favourable warranties and guarantees
  - Need to tender out a contract for the work to be done

- Lack of sufficient funds at the present time may necessitate a phased introduction of a new or revised catering system spread over a given time period (months or years) – as opposed to 'full and immediate' introduction
- Attention must be paid to all costs the enterprise will incur as a result of the introduction of a new or revised catering system – in addition to possible 'opportunity cost' and purchase or leasing costs these may relate to:
  - Lost revenue while kitchen is closed for renovation
  - Cost of removing old systems and equipment – note it may be possible to sell some of the old equipment so this must also be factored into overall considerations and calculations
  - Revisions to SOPs, workplace policies, plans, checklists and standard recipes
  - Training
  - Insurance
  - Service and maintenance.

Staff constraints

In relation to staff (human resources) constraints you will need to consider may include:

- Labour budget for the food production and food service operation – this is always a concern and is traditionally calculated as a given percentage of expected sales.

  The system you decide on must not require staff levels which exceed labour budget parameters
- Labour cost to transport prepared food – in systems where remote kitchens will be used to re-heat food
- Number of skilled staff required – to operate the system: the need for 'skilled' staff introduces potential additional expenses in terms of:
  - Recruitment
  - Remuneration – higher skilled staff attract higher pay rates
  - Training
- Current skill levels of existing staff – in relation to issues such as:
  - Need to recruit additional staff
  - Need to train staff
  - Need to multi-skill employees.

Space constraints

'Space' constraints refer to the amount of room you have available for the new/updated catering system.

Considerations include:

- In most/many cases where there is an existing system you will be expected to fit the new/revised system into the existing space – while it is usually acceptable to use *less* than the current amount of space, it is generally not acceptable or possible to occupy more space
- Use of more space results in added 'opportunity cost' – that is, the loss of the extra space is a cost to the business because it cannot be used to generate income (through setting extra tables to enable the selling and service of extra meals)
- The need to position the food production areas to integrate efficiently and effectively with other stages of the flow of food within the kitchen, in relation to activities such as:
  - 'Goods in' to the kitchen – receival of goods from suppliers
  - Storage of ingredients – in relation to:
    - Dry goods storage
    - Refrigerated storage
    - Frozen storage
  - Food preparation – prior to production/cooking of food
  - Production/cooking – of menu items
  - Post-cooking storage
  - Re-constitution of menu items – where applicable
  - Re-thermalisation – where applicable
  - Distribution to other outlets
- The need to locate food production facilities to facilitate:
  - Food service
  - Access by customers and wait staff.

Compliance constraints

In relation to compliance constraints you need to ensure the catering system/option you choose will:

- Enable compliance with host country food safety legislation
- Complies with all requirements of any HACCP-based Food Safety Plan/Program used by the kitchen – or enables a new HACCP-based FSP to be developed to reflect the new/revised catering system
- Reflect industry 'best practice' in terms of:
  - Food handling practices, procedures, policies and protocols
  - Food production and food service.

Timing constraints

The decision to select, install and commission a new/revised catering system is usually subject to some form of time-related limits.

Considerations may relate to:

- The need to the new or revised system to be fully-functional by a given date – so advanced bookings can be catered for and catering contracts can be fulfilled
- The need for certain stages of the project to be completed by nominated dates – for example, specific timelines may be set for:
  - Research and decision making
  - Releasing Request For Tender (RFT) documentation
  - Awarding the contract
- The need for money to be spent by a given date – in order:
  - For expenditure to qualify for taxation claims
  - That grants are used by the date required.

Existing equipment constraints

Where the project requires you to up-date the catering system in an existing venue, as opposed to introducing a totally new system, there will be constraints in relation to:

- Ensuring new or up-dated equipment integrates with other existing items
- Making sure new technologies are compatible with other technologies which are currently in place
- New equipment will physically fit in the space left when old items have been removed.

Note

It is relatively easier to have to select a catering system for a new venture/enterprise than it is to select a catering system to up-date an existing workplace/business.

When dealing with an existing business several factors have emerged over time as being difficult factors to address – for example:

- There is a general tendency to want to stick to what is known – this often results in (simply) an up-dated version of the previous system being selected.

  In practice the 'old' approach is retained while using more modern equipment
- There is often a reluctance to get rid of some existing items, equipment and practices – the dominant thinking is often:
  - "We cannot afford to do away with that"

    or
  - "We will make do with that item because it still works: we will replace it when it breaks down".

- There is a reduced potential/capacity or predisposition for:
  - Changing the layout of the operation
  - Altering the allocation of space for the system.

When working to select a catering system for a brand new facility there is commonly more scope for flexibility in what can be chosen: the selection of the catering system, of course, should occur as part of the planning process for the new kitchen/business so relevant plans can be prepared to guide construction and installation.

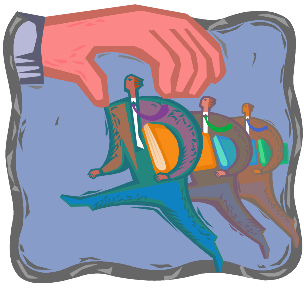
Working to choose a system for a new operation:

- Provides the potential to look at broader opportunities and larger volumes – which can introduce the potential for including satellite kitchens and service points supported by a main/central kitchen
- Allows you to design the total flow of food in the kitchen – from delivery of food into the kitchen, through storage, preparation and cooking, to service and post-production storage and/or distribution to other service points or satellite kitchens
- Generally sees management and other decision makers more well-disposed to new thinking and ideas – in terms of food production options and techniques, different food service strategies and more innovative ways of doing business.

Key Selection Criteria

Key Selection Criteria (KSC) are non-negotiable aspects in relation to selection of a catering system which must be met.

KSC may relate to:



- Catering requirements – for example:
  - The system must be able to produce X meals per session
  - Certain types and styles of food must be able to be produced
  - Nominated standards must be achieved – for example, in terms of:
    - Quality – taste and appearance
    - Nutrition
    - Food safety
- Enterprise constraints – for example:
  - The identified budget must not be exceeded
  - The system must fit within the existing kitchen space
  - The system must be fully-operational by a given date.

All factors identified as KSC must form the basis for:

- Evaluation of catering systems/alternatives being considered – see section 2.2
- Recommendations for a catering system – see section 3.1.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

1.1 Please note: this Work Project forms the basis of Work Project 2.1 and 3.1.

To meet the requirements of this Work Project you are asked to:

- Identify and describe a workplace/enterprise which will form the basis for this and all following Work Projects in this unit
- Research the catering requirements of this workplace/enterprise and provide details of agreed catering requirements
- Research and identify the constraints for the workplace/enterprise which apply to selection of a catering system to meet identified need.

Summary

Establish enterprise requirements for a catering system

When establishing enterprise requirements for a catering system:

- Apply suitable research methods to the process
- Meet and talk with management and operational staff
- Obtain and read published information on systems
- Meet with equipment and system suppliers
- Visit other venues/kitchens
- View/review your own operation
- Determine nature and requirements of the kitchen/venue and available space
- Identify menu items and necessary production levels/volumes
- Specify holding and storage requirements and available space
- Locate service outlets/points as well as available space
- Nominate dietary/nutritional requirements
- Detail power/energy and water needs
- Determine food production stages involved
- Consider existing enterprise standards and practices
- Obtain a mix of data including ‘soft’ and ‘hard’ as well as secondary and primary data
- Involve relevant others in the process
- Identify and quantify all constraints and limitations
- Determine Key Selection Criteria.

Element 2: Evaluate catering systems

2.1 Identify a range of alternative catering systems

Introduction

When selecting a catering system it is best to consider a range of alternatives currently popular with industry operators.

This section discusses the conventional catering system, ready prepared options, commissary service and the assembly/serve system. It also introduces *sous vide*.

System classifications

The catering system classifications/options available today were first described in 1977 by Unklesbay (*Foodservice systems: Product flow and microbial quality and safety of foods*) as:

- Conventional or traditional
- Ready prepared
- Commissary
- Assembly/serve.

Food processing continuum

Unklesbay emphasised the link between food *production* and food *service* using a continuum of food processing to illustrate and explain:

- Some venues buy raw ingredients and prepare all their meals/foods on-site
- Many kitchens buy a combination of raw ingredients, pre-prepared items and ready-made foods and prepare their menus from this mix
- Some service points buy and receive food which is all fully-prepared or pre-portioned – and only need to re-heat or plate it for service with little or no requirement for 'processing'.

In practice the use of pre-prepared and ready-made foods is increasing in many kitchens:

- To save labour costs – because using pre-prepared and ready-made items saves on preparation time
- As the quality of these products continues to improve
- Through the use of detailed food purchasing specifications – detailing how meat is to be trimmed, the size of pieces, the thickness of cuts and the weight of items.

The Conventional system

The ‘conventional’ catering system is ‘cook and serve’.

In this system the food is prepared/cooked and served at the time, either hot or cold depending on type of menu item.

Food is not prepared today for service at a later date.

The food which is processed/cooked may be purchased across all points of the food processing continuum as:

- Raw ingredients – requiring full processing including preparation and cooking
- Pre-prepared food – requiring no or partial preparation prior to cooking/inclusion in menu items
- Ready-made items – only needing to be cut or portion-controlled, plated and served.

This style of service is the most commonly used system and features preparation and cooking of the food at the same location where the food is served.

Food is either:

- Fully cooked to order – as for à la carte service
- Cooked in advance, such as roasts and wet dishes, and held hot (60˚C or above) – ready for service
- Prepared in advance, such as ice creams, cold entrées and other cold desserts, and held cold (at or below 5˚C).

Individual kitchens/venues will have house policies regarding:

- Quantity of food to be prepared – based on expected demand
- Treatment of left-over food – for example:
  - Hospitals and aged care facilities commonly have a policy stating no left-over food is to be stored for later re-use to optimise food safety and avoid the dangers inherent in storing and re-heating previously cooked food
  - Hotels and restaurants may allow left-over food to be stored providing:
    - It is properly labelled
    - It is correctly stored
    - It is used within three days
    - It is re-heated correctly
    - It is only re-heated once.

The food preparation and cooking equipment found in a kitchen using the conventional catering approach is very diverse, reflecting the cooking style and methods of dishes listed on the menu.

Centralised and decentralised service

Use of a Conventional catering system can be applied to operations where the service of the food is:

- Centralised – that is, food service occurs at or adjacent to the food production area
- Decentralised – that is, where the food is transported by tray, trolley, conveyor belt, to other location within the same business where it is either plated or served.

Ready prepared

The ready prepared catering system focuses on preparing food on-site, storing it on-site, under refrigeration or frozen storage, and then re-heating it on-site, when required, for on-site service.

As with the Conventional system the food which is processed or cooked may be purchased across all points of the food processing continuum as:

- Raw ingredients – requiring full processing (preparation and cooking)
- Pre-prepared food – requiring no or partial preparation prior to cooking/inclusion in menu items
- Ready-made items – only needing to be cut or portion-controlled, plated and served.

The options in this classification are:

- Cook-chill
- Cook-freeze.

Cook-chill

Cook-chill is a system which has six stages:

- Produces cooked food – by standard or bulk-cooking methods such as the use of kettles or cook tanks
- Packages cooked food using an automated pump system or a manual-filling option [hand-held jug or ladle] – in a variety of containers but commonly into a range of durable, vacuum-sealed plastic bags catering for individual serve sizes up to larger volume/bulk packs
- Rapidly chills cooked food – using blast chilling, ice slurry tumblers or iced water bath causing food to reduce from cooking temperatures to 5˚C or less in 90 minutes

- Stores the food under controlled refrigerated conditions – in the range of -2˚C to 0˚C for periods of up to seven weeks
- Re-heats the food as required – using options including:
  - Steamers
  - Braising pans
  - Microwave ovens
  - Kettles
  - Combi-ovens
- Holds the food for plating and service.

Cook-freeze

Cook-freeze is a system which has seven stages:

- Produces food which is 'almost cooked' – by standard or bulk-cooking methods (such as the use of kettles or cook tanks)
- Packages cooked food using an automated pump system or a manual-filling option – in a variety of containers but commonly into a range of durable, vacuum-sealed plastic bags catering for individual serve sizes up to larger bulk packs
- Rapidly freezes cooked food – using blast freezers causing food to reduce from cooking temperatures to -20˚C or less in 90 minutes
- Stores the food under controlled freezer conditions – in the range of -20˚C for months
- Requires thawing of frozen product (to 0˚C to 4˚C) prior to re-heating – under strict time-temperature controlled conditions
- Re-heats the food as required – using options including:
  - Steamers
  - Braising pans
  - Microwave ovens
  - Kettles
  - Combi-ovens
- Holds the food for plating and service.

Commissary

The Commissary catering system is one where food:

- Is produced in bulk in a central kitchen
- Is then distributed, usually hot or cold but may be frozen, to satellite kitchens or *commissaries* remote from the main kitchen – food may be transported:
  - In bulk
  - Portion-controlled (individual/single serves) – pre-plated for service.

Satellite kitchens to which food is transported may be:

- Relatively close
- At a significant distance.

When the food arrives at the satellite kitchens it may be:

- Served immediately
- Stored – under refrigeration or in freezers, and re-heated as necessary.

Traditionally the satellite kitchens require little or no food equipment apart from

- Re-heating units
- Food display and service equipment – to assist with food service.

Assembly/serve

The assembly/serve system is not commonly suitable for commercial outlets.

It features:

- Purchase of prepared dishes/menu items – from suppliers
- Storage of pre-prepared items on the premises – as appropriate to each item. under:
  - Refrigeration
  - Frozen storage
- Only basic food activity in relation to the pre-prepared menu items – such as:
  - Portioning
  - Plating
  - Re-heating
  - Service.

Sous vide

Two definitions of *sous vide* will assist understanding of this process:

"The term *sous vide* means *under vacuum* and describes a processing technique whereby freshly prepared foods are vacuum sealed in individual packages and then pasteurised at time-temperature combinations sufficient to destroy vegetative pathogens but mild enough to maximise the sensory characteristics of the product." ('*Cook chill for foodservice and manufacturing: Guidelines for safe production, storage and distribution*', Cox and Bauler, 2008, p.201).

"*Sous vide*: a process of sealing raw, fresh food items in plastic pouches to allow chilled storage and then cooking in boiling water prior to service." ('*Foodservice Organisations: A managerial and systems approach*', Gregoire, 2010, p.76).

The *sous vide* approach:

- Requires raw ingredients to be stored under refrigeration – and to be cold (3˚C) when vacuum sealed
- Vacuum packs raw food into individual bags – at different pressures depending on the type of product: for example delicate fish would be packed at a lower pressure than a hard, root vegetable
- Demands vacuum packed food is either immediately:
  - Cooked and served
  - Stored at or below 1˚C
- Cooks the food in its plastic bag using an immersion circulator – operating at lower than normal temperatures (for example, 60˚C to 65˚C) but for longer periods thereby producing a better quality result
- Is regarded by most as an adjunct to traditional food production options – more so than as a totally *alternative* system across their entire menu.

  This means kitchens that use *sous vide* will use it for certain dishes they believe respond well to the approach, but not for all menu items.

Combination approach

In some cases the kitchen may choose to use a mix of Ready prepared foods and cook-chill, cook-freeze or *sous vide items* food – for example:

- The primary ingredient on the plate may be cook-chill
- The vegetables may be cooked that day/for the session
- The sauce may be cook-chill
- Certain menu items may be cook-freeze
- Other dishes may be *sous vide*
- Some menu items may be fully-prepared and ready-to-serve.

Online information

Visit the following sites for more information:

Kitchen equipment

- http://www.youtube.com/watch?v=cHxpZWS5Ogg 'Commercial Kitchen Equipments' (6 minutes 1 second)
- http://www.youtube.com/watch?v=s409Agd6rkI 'Kitchen Equipment' (1 minute 7 seconds)
- http://www.youtube.com/watch?v=y97Gee05Cco 'Joni Steam Jacketed Cooking Kettles: Mince Beef' (1 minutes 34 seconds)

Cook-chill

- http://www.youtube.com/watch?v=jAbFCLNnpvc 'How to Cook-Chill: Cook-Chill Technology (1 minute 57 seconds)
- http://www.youtube.com/watch?v=8qJtWN5Tr8w 'The Cook Chill Process' (8 minutes 12 seconds)
- http://www.youtube.com/watch?v=nnsPHhuhk90 'Regethermic Cook Chill System' (9 minutes 46 seconds)
- http://www.youtube.com/watch?v=8yLZ9EpwO3w 'Cook Chill Food System: D C Norris & Company Ltd (7 minutes 33 seconds)
- http://www.youtube.com/watch?v=kYbNszCCi1g 'The Cook-Chill Process' (8 minutes 18 seconds)

Sous vide

- http://www.youtube.com/watch?v=5PbnRF-ePnI 'Tucs combination chiller-cook chill sous vide processor' (4 minutes 23 seconds)
- http://www.youtube.com/watch?v=5WuHddsoV70 'Sous vide lamb rack with infused reduction sauce & sautéed vegetables' (8 minutes 45 seconds)
- http://www.youtube.com/watch?v=Rl-msqKc_YQ 'Sous vide' (6 mins 41 seconds)

2.2 Evaluate agreed enterprise requirements against systems

Introduction

It is essential to evaluate the catering systems being considered against the requirements and constraints identified in the planning process.

This section highlights the role of Key Selection Criteria, provides an overview of the evaluation process, addresses factors to consider when evaluating catering system options for selection and identifies the potential for needing to compromise.

Role of KSC

The Key Selection Criteria must form the basis of all evaluations.

This highlights the need to:

- Comprehensively establish these criteria at the beginning of the catering selection research process – so there is certainty about what is required
- Know and understand what the specifics of these criteria are
- Keep these KSC central to all considerations and analysis.

The evaluation process

Evaluation is a process of comparison.

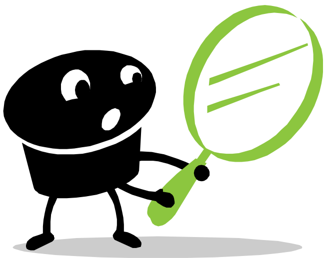
The evaluation process comprises three elements which:

- Compares what is available against what is required – through asking a series of relevant questions
- Judges the degree to which there is alignment between requirements, constraints and availability
- Determines the relative advantages and disadvantages of available options.

Keys in undertaking an evaluation are:

- Consider all relevant factors – not just one or two, or 'most' of them
- Allocate sufficient time for evaluation – never rush this stage of the process
- Use a team of people to undertake the evaluation – as opposed to doing it on your own
- Document your thoughts and findings – never rely solely on memory to provide a foundation for discussion and decision making.

Factors to consider



A combination of the following topics is commonly used as a matrix against which catering system options can/should be evaluated.

The exact nature of the questions will vary between properties but those presented below are indicative.

Evaluation should determine:

Nature of the operation

- Whether the system suits the *type of operation* being considered:
  - Is the option going to be effective in the context it is being expected to operate?
  - Is it acceptable to the customers?
  - Does it enable the operating and performance targets/goals to be achieved?
  - Are other industry leaders in the sector using this option?
  - Does a system give the business potential to move into another market/niche?
- Whether the system aligns with the *market position* of the operation:
  - Does the system fit with public image of the business?
  - Will the system give the business a competitive advantage/USP which would be useful in the future?
  - What do customers think of the system and the food it produces?

The menu

- Whether the *intended menu* can be produced:
  - Is the system appropriate to the food or quality standard required?
  - Can all identified menu items be produced and served using the system?
  - Does the system have the capability to expand to offer and produce other types of dishes, cooking styles or cooking methods?
  - What new menu items or food types can be offered or produced using a certain system?

Production volume

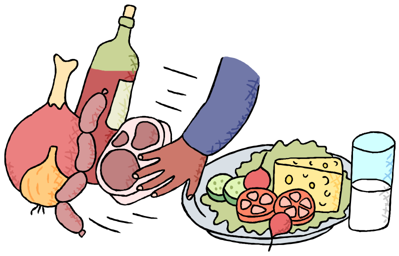
- Whether the system has the required *capacity*:
  - Will the system be able to cope for identified peak demands on an hourly, sessional or daily basis?
  - Does it have the potential to be supplemented by extra equipment to enable capacity to grow if demand increases?
  - Will the system deliver production volume requirements for events/functions, as well as cater for other known demand at the same time?
  - What new/extra capacity does a system bring to the kitchen?

Service areas/service points

- Whether *food service* potential meets requirements:
  - Can the menu items be served in accordance with operator standards and at the required service outlets?
  - Will required service numbers be achieved in the given timeframes for service?
  - Does food service reflect identified customer preferences for service?
  - Are suitable food transportation facilities or resources available to support the system?
  - Does a system open up new opportunities for fresh service points?

Storage and holding

- Whether *pre-production storage* is sufficient to meet expected demand:
- Are dry goods stores, cool rooms and freezers available as required?
- Are they of sufficient size?
- Are they located to facilitate the flow of goods/foods within the kitchen/food production area?
- What extra capacity does a new system offer?
- Whether *post-production storage* is sufficient to meet expected demand:
  - Is there capacity to hold sufficient hot food for service and storage, as required?
  - Is there capacity to hold sufficient cold food for service and storage, as required?
  - Is there capacity to *display* hot and cold food, as required?
  - What extra capacity does a new/alternate system offer?



Nutritional and dietary requirements

- Whether the food produced meets guidelines for *nutritional content*:
  - What proof is there the required nutritional content can be obtained?
  - What additional nutritional content can be attained using certain systems?
  - How is this data obtained and who generates it?
  - What on-going checks can be done to verify nutritional content?
- Whether the system allows the production of identified menu items to enable provision of identified diet-specific meals:
  - What menu items for specific diets can be produced?
  - What extra menu items can be produced using certain systems?

Available space

- Whether the system fits the *allocated space*:
  - Do all elements of the entire food production system fit into the space available for all stages of food production - receiving, pre-production storing, preparation, cooking, post-production storing?
  - Do all elements of the entire food production system fit into the space available for all stages of food service - service, holding, display, distribution?
  - What extra space is required?
  - What space saving has been made?

Ingredients purchased

- Whether the system impacts the *ingredients purchased* by the venue :
  - How does each system impact the current way the kitchen orders or purchases ingredients?
  - Will more fresh food/ingredients be required?
  - Will more ready-made foods be required?
  - How does this fresh-versus ready-made dichotomy fit with quality standards and customer perceptions?

Enterprise practices and standards

- Whether systems will impact enterprise *practices and standards*:
  - Will existing kitchen SOPs need to change?
  - If so, which ones and to what extent? Is this change acceptable?
  - Will existing standards need to change?
  - If so, which ones and to what extent?
  - What *new* practices, standards and organisational change will need to be introduced?

Logistics

- Whether the system is a *realistic and feasible* option:
  - Can required base materials (food/ingredients suitable to the process) be delivered on a reliable, on-going basis?
  - Can sufficient numbers of suitable qualified staff be recruited to operate the system or be trained in-house?
  - Are the utilities required to support and operate the system available?
  - Are cost prices for utilities acceptable or viable?
  - What installation requirements apply to a new system?
  - Can transportation be safely achieved on a regular on-going basis, if required?
  - Who will supply, install and commission the system and system elements?
  - What is known about them and what guarantees and warranties do they provide?

Financial constraints

- Whether the system aligns with the identified *financial parameters* for the initiative:
    - How much do suitable options cost?
    - What quotations were obtained?
    - What discounts are available?
    - How did prices for similar equipment/technology differ between different manufacturers or suppliers?
    - What financing is available to support the acquisition, and what is the cost of it?
    - What existing equipment and facilities can be used or re-used as part of the new system?
    - What revenue can be generated through the sale of equipment being removed to make way for a new system?

Compliance requirements

- Whether the system/s meet identified *compliance requirements*:
    - Is the system compliant with legislated food safety obligations?
    - Will the system enable implementation of a HACCP-based FSP?
    - Will a new FSP be required or will the existing one remain applicable?
    - Does the system conform to any applicable industry or other relevant COPs?
    - Can the system support implementation of QA as required by the operator?

Timing constraints

- Whether the preferred system can be introduced in accordance with necessary timing limitations:
    - How long will it take to remove the existing equipment and systems or technology from the food production and service areas?
    - What impact will this have on trade, cash flow and profit or business viability?
    - How long will it take to install and commission new systems or technology and equipment?
    - Does installation and commissioning align with identified operational requirements, or does it create an operational problem in meeting projected demand?

Operating costs

- Whether the system is viable in terms of:
  - What is the cost of staff training to bring staff up-to-speed for the system in all locations?
  - Are the projected maintenance costs for the system viable?
  - What potential wastage is predicted and how does this compare to current wastage levels?
  - How much energy will the system use and at what cost?

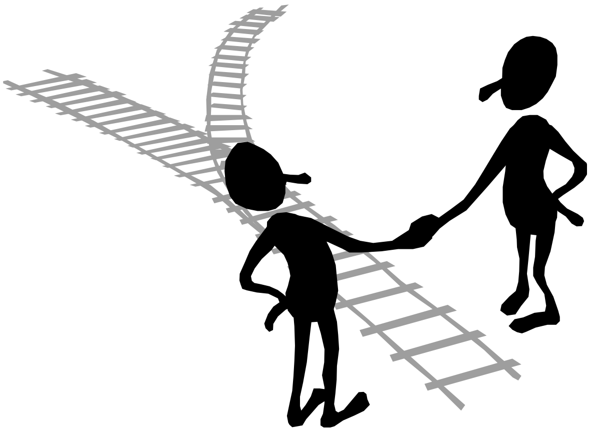
The possible need to compromise

Compromising means settling for something less than what was originally needed or wanted.

While KSC must always form the basis for evaluating options the realities of business life will nearly always require you to compromise.

This may mean you have to:

- Extend installation and commissioning dates for equipment beyond preferred timelines
- Spend more on the system you will decide to acquire
- Use more space than intended to house the system which best serves your needs
- Buy from a supplier or manufacturer you did not originally want to deal with
- Change enterprise policies, procedures or protocols you wanted to remain the same.



Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

2.1 Please note: this Work Project flows from Work Project 1.1 and forms the basis for Work Project 3.1.

To meet the requirements of this Work Project you are required to:

- Identify and describe at least two catering system options which could be theoretically selected to suit the listed catering requirements and enterprise constraints of the workplace/enterprise used as the basis for Work Project 1.1
- Evaluate identified catering requirements and enterprise constraints against both of the above catering system options.

Summary

Evaluate catering systems

When evaluating catering systems:

- Identify and research all systems relevant to identified catering requirements and enterprise constraints
- Understand the impact/importance of the food processing continuum on various systems
- Become familiar with the Conventional system
- Know the difference between 'centralised' and 'decentralised' service
- Differentiate between cook-chill and cook-freeze options
- Be able to describe the commissary option using satellite kitchens
- Understand the assembly-serve system
- Note the options provided by the *sous vide* system
- Realise an effective system may use a combination of different approaches
- Use Key Selection Criteria and all other relevant factors (including identified constraints) as basis for evaluating catering systems
- Involve others in the evaluation process
- Ask lots of questions
- Document thoughts and findings of this stage of the process.

Element 3:
Recommend a catering system

3.1 Consider the advantages and disadvantages of systems in making recommendation

Introduction

When relevant catering systems have been identified and evaluated it next remains to make recommendations based on the research and analysis which has been undertaken.

This section identifies the basis for making recommendations, presents an overview of advantages and disadvantages of various systems as well as ancillary topics for consideration and describes possible activities in the recommendation procedure.

Basis for making recommendations

When making recommendations for the selection of a catering system:

- Involve relevant stakeholders in the process – do not do it on your own
- Base your recommendations on facts identified during the research process – which must reflect the identified requirements and constraints established at the start of the process
- Be sure to distinguish and make completely clear any aspects of the report which are thoughts and opinions – as distinct from 'fact' information
- Include details of all classifications of data collected as part of the research/investigative process – qualitative and quantitative data, as well secondary and primary data
- Make a definite recommendation – detailing:
  - Name/type of system
  - Manufacturer's name
  - Equipment and technology to be used
  - Dates for strategic action – such as:
    - Preparing the site/removing existing equipment
    - Installing and commissioning equipment and systems
    - Trailing equipment and systems
    - Supporting/ancillary action – development of new or revised menus and SOPs; advertising; staff training
  - Costs – including deposits required and dates progress payments, acquisition options, funding options and costs, service and maintenance
- Prepare a written report – and distribute to the decision makers in the organisation

- Organise a meeting where you present your recommendations – and:
  - Explain your recommendations
  - Justify your conclusions
  - Include a Question and Answer session.

Overview of advantages and disadvantages of various systems

Depending on individual venue catering requirements and specific organisational and operational constraints you may illustrate your recommendations with reference to the following points as appropriate to the system being considered.

Advantages of the Conventional system

Commonly held thoughts on the advantages of the Conventional catering system are:

- It produces food of a high quality – but this, of course, is always tied to many other factors too such as quality of initial ingredients, staff skills and expertise, recipes, and available time.

  The point being using the conventional system is not a guarantee of high quality food but, generally speaking/all other things being equal it is regarded as producing better quality food than other options – 'fresh is best'

- The public/customers are well-disposed towards traditional kitchens which cook and serve food in this way – they:
  - Have confidence in the quality and safety of the food
  - Prefer fresh-cooked to other forms of food production/service
  - Appreciate venues where the operations of the kitchen are visible from the dining area
- Most cooks, chefs and kitchen staff are familiar with the operations of a conventional kitchen – so this means:
  - A wider pool of trained and experienced staff to choose from when recruiting/selecting staff
  - Less need to train staff – because they are more likely to be familiar with and competent in what is required
- The system provides opportunity to be more flexible and responsive to immediate need – for example:
  - The kitchen can quickly respond to a new (or cut-price) food which becomes available – and have a new dish on the menu and available for service literally within hours
  - Cooks can cater for special requests from customers on-the-spot, or with very short lead times
- Holding or refrigeration or freezer space for food is minimised – there is primarily a need to only store food prior to preparation or production and not after it has been produced
- The equipment available in an existing conventional kitchen can often be used to prepare a large variety of different menu items – even when the menu changes there may not be a need to purchase new equipment.

Disadvantages of the Conventional system

While the advantages of the Conventional system are substantial and important there are, nonetheless, significant disadvantages to this approach:

- Higher foods costs per unit produced – the cooked-to-order approach cannot achieve the economies of scale available through other bulk-food approaches to catering
- Consistency of quality is sometimes an issue – as individual dishes can result in an unacceptable level of quality in terms of taste, aroma, and appearance
- Higher labour costs – the cooked-to-order nature of many foods produced under the conventional system, as well as the traditionally higher costs associated with food preparation prior to cooking – trimming, peeling, cutting, portioning, requires more staff time which translates into more wages needing to be paid
- Potential need for extra equipment – while the menu should always dictate the equipment needed in a kitchen the conventional kitchen commonly requires a greater variety of items of equipment, meaning:
  - Potentially higher initial equipment and on-going running costs
  - Possible need for a greater kitchen space to accommodate all of the required equipment
- Potential for reduced food safety – while this is not necessarily a significant concern in all conventional systems it can occur where:
  - Left-overs occur at the end of food service and they are not stored properly and/or and re-heated
  - There are inadequate controls over the wide variety of food activities which take place in traditional conventional kitchens and food service areas.

Advantages of Ready prepared systems

Advantages attributed to the Ready prepared systems come from the reasons the systems were introduced in the first place as well as the benefits flowing from them:

- Reduced costs – in terms of:
  - Lower labour cost per unit produced due to economies of scale – meaning a menu item for multiple days of service can be prepared and produced at the one time and then held for (re-heating and) service, as required
  - Economies of scale related to:
    - Bulk buying of ingredients
    - Energy usage

  - Lower levels of wastage – as:
    - Left-overs are virtually eliminated
    - Only required food is removed from refrigerated and frozen inventory thus eliminating over-production by directly matching the type and quantity of food which is re-heated to orders

- Better yield from food items – as a result of the cooking processes used which can significantly decrease shrinkage and loss usually attached to other conventional cooking methods
- Addresses shortages of skilled labour – through using a system consistent with employing a lower-skilled labour force, compared to hiring highly trained and experienced Chefs
- Produces food of a consistent quality or standard – the techniques, technology and controls make for extremely high levels of consistent quality end product
- Food service can occur at any time – because the food is already available. It just needs to be re-heated and served.

  This provides enormous flexibility in service times and enhances the ability to serve food 'on demand'.

Disadvantages of the Ready prepared system

Several significant disadvantages attach to the Ready prepared option and these have seen its use restricted to institutional catering.

The disadvantages are seen as:

Adverse customer reaction– where it is known a business operates this style of service there is frequently a negative backlash from patrons

If you are operating in an environment where the consumers have no real choice/option, this may not be a concern but where you operate in an environment rich with more-preferred catering options it will be a major consideration in your decision-making

- Decreased levels of food quality – this is regarded by some as a 'fact' in relation to food produced and served under this system, and regarded by others simply as a customer 'perception'.

  Debate continues regarding the actual levels/standards of food quality in relation to Ready prepared food but the following points appear constant:

  - The quality of dishes served under this system is improving over time
  - When questioned customers consistently say they prefer food produced via the Conventional system

- Increased establishment costs – there will be a need for higher levels of expenditure when setting up this system to cover:
  - Larger requirements for storage, either refrigerated and frozen, of prepared food
  - Specialist equipment and utensils and food area for the packaging of food ready for storage following production
  - Equipment required for re-thermalisation of refrigerated and frozen menu items
- Potential for limited or restricted menu choices – it is a fact of cooking life that certain dishes do not 'hold' or re-heat well.

Their appearance, texture and overall quality can be dramatically compromised by holding and re-heating meaning these menu items are unsuitable for this catering system option and can or should not be offered.

Another limiting factor is some cook-chill systems can only process foods which can be 'pumped' into storage packaging, limiting their application to sauces, soups, dressings, stews and casseroles, gravy, pie fillings and mashed potatoes.

Note: some point to the fact the Ready Prepared approach can increase menu choices through holding a wide variety of dishes, under frozen storage, on a single-serve basis and re-heating single units as they are ordered.

Proponents of this line of thought say the same variety of menu items would not be possible under a Conventional system

- Higher loss or other 'damage' potential from an out-of-control food handling event – for example:
  - An error or problem when producing or cooking an item will result in greater loss of product due to the higher volumes of food being handled at any one time
  - If a batch of food is contaminated it will affect more people because larger quantities are being prepared at the one time
- Need to hire staff with specialist skills or train staff in work roles, including tasks and activities, not normally undertaken as part of standard roles and responsibilities under Conventional catering system – such as:
  - Vacuum sealing food in bags
  - Operating pasteurising and chilling equipment
  - Testing pH levels
  - Preparing and attaching suitable labels to packages placed into storage
  - Monitoring and controlling times and temperatures of food prior to, and during, refrigerated or frozen storage
  - Operating re-thermalisation equipment.

Advantages of the Commissary system

Advantages include:

- Central control of quality and standards
- Reduced need for skilled staff in satellite kitchens
- Staff at the main kitchen will operate at high levels of productivity – helping reduce costs per unit
- High volume will bring a range of 'economies of scale'
- No/little need for food processing or cooking equipment in satellite kitchens – may only be a need for storage, re-heating and service equipment
- Facilitates the operation of multiple outlets and new service points

- Allows main kitchen to be built in an area where land and building costs are most competitive – as opposed to buying land and building a central kitchen in high-value geographical location.

Disadvantages of the Commissary system

Disadvantages include:

- Quality-related issues – associated with:
  - Customer perceptions about loss of quality because food is not 'freshly cooked'
  - Actual decrease in food taste, appearance, food safety and nutritional content
- Only food produced by central kitchen – or foods bought-in 'fully prepared'– can be offered
- Satellite kitchens cannot respond to individual customer demand or preferences – food is essentially presented on a 'take-it-or-leave-it' basis
- Main kitchen requires highly-skilled, highly-competent staff
- Requires expenditure on:
  - Transporting the food – such as food transport vehicles and hot and cold food carts
  - Special packaging for foods
- Need for food safety protocols to be developed to cover transportation of food – especially relating to:
  - Time-temperature controls
  - Protection of food from contamination
  - Delivery schedules
  - Servicing of the food transport vehicle
  - Actions of the delivery driver.

Advantages of the Assembly/Serve system

The advantages of this approach are:

- Low levels of equipment required
- Less space required
- Reduced labour cost – due to:
  - Lower levels of skilled staff required
  - Fewer staff or hours required to provide labour
- Service can be provided 'at any time'
- Service is usually relatively prompt.

Disadvantages of the Assembly/Serve system

The disadvantages of this approach are:

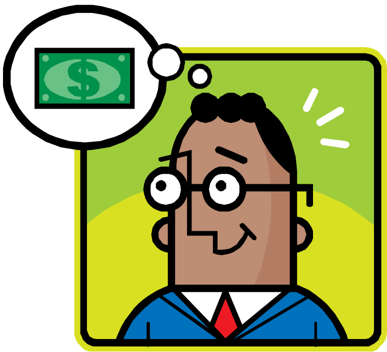
- Limited choice of menu items – the outlet can only offer items available from suppliers
- Totally reliant on what suppliers can offer or deliver
- Relatively high food cost – because all menu items are 'bought in' as opposed to being prepared from less expensive raw ingredients
- No capacity to respond to individual customer need and any special requests

- Quality-related issues – associated with:
    - Customer perceptions about loss of quality because food is not 'freshly cooked'
    - Actual decrease in food taste, appearance, food safety and nutritional content.

Ancillary topics for consideration

In addition to the advantages and disadvantages identified above there will commonly also be a need to address:

- Calculation and comparison of costs of outright purchase against other acquisition options (such as leasing) factoring in relevant issues such as:
    - Taxation implications
    - Depreciation rates
    - Cash flow
    - Cost-Benefit analysis



- Consideration of strategies for dealing with equipment breakdown, maintenance and service/repair needs – such as:
    - Service contracts with external providers
    - Maintaining replacement items and parts
    - Establishing and maintaining an in-house Maintenance department
- Comparison and contrast of:
    - Curren situation to expectations
- Costs and timing related to:
    - Removing existing equipment
    - Installation of new equipment, systems and technology
    - Commissioning and testing the system
- Organisational changes required – such as:
    - Requirements for new or revised:
        - Policies and SOPs
        - Documentation to support QA system and FSP requirements
        - Menus
        - Service times
        - Staff rosters
        - Ordering and purchasing of raw materials and ingredients
        - Stock control and management
        - Internal communication
    - Need for new or different staff training and induction and orientation for staff

- Changed arrangements for:
  - Determining productivity
  - Yield testing
  - Accessibility to raw materials and end-products.

Possible activities in the recommendation procedure

The following activities may assist in delivering your recommendation to management/the decision makers in the organisation:

- Identify a date, time and venue for the presentation of your recommendations
- Invite all relevant stakeholders to the meeting – urge them to attend by stressing the importance of the meeting
- Circulate your written report prior to the presentation – so stakeholders have an opportunity to read the material, become familiar with the content, formulate questions they want to ask and identify issues they want to raise

- Include a tour of a venue already using the catering system you are recommending – so stakeholders obtain first-hand experience of the system in terms of:
  - Inspecting facilities and equipment
  - Identifying flow of food through the venue/kitchen
  - Observing food preparation and production
  - Watching food service
  - Experiencing the food produced
- Invite managers and operators from a kitchen using the system you are recommending to attend the meeting – to give their endorsement and opinions of the system
- Ask suppliers of the system to attend – and:
  - Speak to your recommendation
  - Provide additional detail relating to the proposal
  - Screen relevant videos/DVDs.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

3.1 Please note: this Work Project flows from Work Projects 1.1 and 2.1.

To meet the requirements of this Work Project you are required to:

- Submit a report making a recommendation for a catering system for the workplace/enterprise identified in Work Project 1.1
- Ensure the report considers the advantages and disadvantages of system options/alternatives.

Summary

Recommend a catering system

When recommending a catering system:

- Involve others in the process
- Base recommendations on facts
- Prepare and circulate a formal, written report
- Organise a meeting to support the written report and to present and defend/explain recommendations
- List relevant and relative advantages and disadvantages of catering systems which have been considered/researched
- Compare different catering systems against each other as well as identified catering requirements and organisational constraints.

Presentation of written work

1. Introduction

It is important for students to present carefully prepared written work. Written presentation in industry must be professional in appearance and accurate in content. If students develop good writing skills whilst studying, they are able to easily transfer those skills to the workplace.

2. Style

Students should write in a style that is simple and concise. Short sentences and paragraphs are easier to read and understand. It helps to write a plan and at least one draft of the written work so that the final product will be well organised. The points presented will then follow a logical sequence and be relevant. Students should frequently refer to the question asked, to keep 'on track'. Teachers recognise and are critical of work that does not answer the question, or is 'padded' with irrelevant material. In summary, remember to:

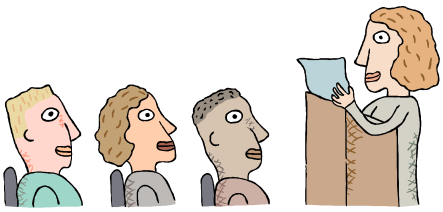
- Plan ahead
- Be clear and concise
- Answer the question
- Proofread the final draft.

3. Presenting Written Work

Types of written work

Students may be asked to write:

- Short and long reports
- Essays
- Records of interviews
- Questionnaires
- Business letters
- Resumes.



Format

All written work should be presented on A4 paper, single-sided with a left-hand margin. If work is word-processed, one-and-a-half or double spacing should be used. Handwritten work must be legible and should also be well spaced to allow for ease of reading. New paragraphs should not be indented but should be separated by a space. Pages must be numbered. If headings are also to be numbered, students should use a logical and sequential system of numbering.

Cover Sheet

All written work should be submitted with a cover sheet stapled to the front that contains:

- The student's name and student number
- The name of the class/unit
- The due date of the work
- The title of the work
- The teacher's name
- A signed declaration that the work does not involve plagiarism.

Keeping a Copy

Students must keep a copy of the written work in case it is lost. This rarely happens but it can be disastrous if a copy has not been kept.

Inclusive language

This means language that includes every section of the population. For instance, if a student were to write 'A nurse is responsible for the patients in her care at all times' it would be implying that all nurses are female and would be excluding male nurses.

Examples of appropriate language are shown on the right:

Mankind	*Humankind*
Barman/maid	*Bar attendant*
Host/hostess	*Host*
Waiter/waitress	*Waiter or waiting staff*

Recommended reading

2012; Robotics and automation in the food industry: Current and future technologies, Woodhead

Australian Cook Chill Council Inc & Australian Cook Chill Council; 2000; Guidelines for chilled food production systems including food safety programs, Australian Cook Chill Council Inc

Cox, Brigitte & Bauler, Marcel; 2008; Cook chill for foodservice and manufacturing: guidelines for safe production, storage and distribution, Australian Institute of Food Science and Technology

Drysdale, John A; 2010; Restaurant food service equipment; Prentice Hall

Food Safety Authority of Ireland; 2006; Cook-chill systems in the food service sector; Food Safety Authority of Ireland

Ghazala, S; 1998; Sous vide and cook-chill processing for the food industry, Aspen Publishers

Greene Belfield-Smith & Trent Regional Health Authority; 1988 (2nd edition); The cook chill system: an appraisal of equipment and consumables; Greene Belfield-Smith

Lieux, Elizabeth McKinney & Luoto, Patricia Kelly; 2008 (3rd edition; , Exploring foodservice systems management through problems; Pearson Prentice Hall

Light, N. D. (Nicholas D.) & Walker, Anne; 1990; Cook-chill catering: technology and management, Elsevier Applied Science

McSwane.D, Linton.R, Rue.N; 2004 (4th edition); Essentials of Food Safety and Sanitation; Prentice Hall

Mortimore, Sara; 2013, HACCP: A practical approach, Springer, New York

National Restaurant Association; (2006, 6th edition); ServSafe Manager; Prentice Hall

Pulle, Mervyn; 2003; Food hazards: factors that affect food safety; Knowledge Books and Software

Scanlon.N; 2012 (4th edition); Catering Management; Wiley

Sheard, Mike & Church, Ivor & Leeds Polytechnic & Sous Vide Advisory Committee 1992, Sous vide cook-chill, Leisure and Consumer Studies,Leeds Polytechnic, Leeds

Sun, Dawen 2012, Handbook of food safety engineering, Wiley-Blackwell, Oxford

Thomas, Chris & Hansen, Bill & Hansen, Bill. Off-premise catering management 2013, Off-premise catering management, 3rd ed, Wiley, Hoboken, N.J

Trainee evaluation sheet

Select catering systems

The following statements are about the competency you have just completed.

Please tick the appropriate box	Agree	Don't Know	Do Not Agree	Does Not Apply
There was too much in this competency to cover without rushing.	☐	☐	☐	☐
Most of the competency seemed relevant to me.	☐	☐	☐	☐
The competency was at the right level for me.	☐	☐	☐	☐
I got enough help from my trainer.	☐	☐	☐	☐
The amount of activities was sufficient.	☐	☐	☐	☐
The competency allowed me to use my own initiative.	☐	☐	☐	☐
My training was well-organised.	☐	☐	☐	☐
My trainer had time to answer my questions.	☐	☐	☐	☐
I understood how I was going to be assessed.	☐	☐	☐	☐
I was given enough time to practice.	☐	☐	☐	☐
My trainer feedback was useful.	☐	☐	☐	☐
Enough equipment was available and it worked well.	☐	☐	☐	☐
The activities were too hard for me.	☐	☐	☐	☐

The best things about this unit were:

The worst things about this unit were:

The things you should change in this unit are:

Trainee Self-Assessment Checklist

As an indicator to your Trainer/Assessor of your readiness for assessment in this unit please complete the following and hand to your Trainer/Assessor.

Select catering systems

		Yes	No*
Element 1: Establish enterprise requirements for a catering system			
1.1	Research catering requirements the enterprise requires	☐	☐
1.2	Identify the enterprise constraints in selecting a system	☐	☐
Element 2: Evaluate catering systems			
2.1	Identify a range of alternative catering systems	☐	☐
2.2	Evaluate agreed enterprise requirements against systems	☐	☐
Element 3: Recommend a catering system			
3.1	Consider the advantages and disadvantages of systems in making recommendation	☐	☐

Statement by Trainee:

I believe I am ready to be assessed on the following as indicated above:

Signed: ________________________________ **Date:** ______ / ______ / ______

Note:

For all boxes where a **No*** is ticked, please provide details of the extra steps or work you need to do to become ready for assessment.

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